PROJECT REPORT

# Object-Context Anomaly Detection

Using CLIP and Hopfield Memory


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| Project at a Glance | |
|---|---|
| Task | Flag detections that may be out of place in an image. |
| Dataset | 551 annotated images across five object classes. |
| Model | YOLO11s-OBB, CLIP ViT-B/32, trained context selector and Hopfield memory. |
| Evaluation | Full inference on 20 validation images, with an additional demo image. |
| Code | context_anomaly.py |


## Abstract

An object detector can find a toothbrush or a screwdriver, but its class prediction says little about where the object belongs. We used a trained YOLO detector to find objects and a small attention model to pick a surrounding crop. CLIP embeds the crop and class name. We retrieve a vector for each from a bank of example contexts and compare them using cosine similarity. We ran the whole pipeline on 20 images. The detector recovered 3 of 22 labelled objects at input size 896 and 16 at input size 256 using a same-class match and polygon IoU of at least 0.50. We report the context scores as observations because the dataset contains only object labels.


## 1. Project Overview and Objectives

The aim was to check whether the surroundings provide a second signal for judging an object detection. We wanted the program to score each detection, show the selected context, and mark low-scoring boxes on the image. We used the saved project weights so the inference can be repeated without retraining.

## 2. Dataset Details

The five classes are screwdriver, comb, knife, pen and toothbrush. The detector labels give four corners for each oriented box. The training split contains 11 images with empty label files.

| Split | Images | Oriented boxes |
|---|---|---|
| Training | 531 | 882 |
| Validation | 20 | 22 |
| Total | 551 | 904 |

**Dataset access.** The supplied *Hopfield networks.zip* contains *dataset.zip* and the unpacked *dataset/* folder. The archive also contains 608 context images in *data/* and 966 in *data_uniform/*. The context collections have no anomaly labels or documented independent splits. A public download URL was unavailable.

## 3. Model Details

**Object detection.** We used the five-class *train4/weights/best.pt* checkpoint. Its training configuration records 30 epochs, batch size 16 and a requested image size of 240. Each detection supplies a class name, confidence and four box corners.

**Context selection.** For multiple detections, we generate four views per object: a column, a row, a crop roughly twice the box size, and the whole image. The selector uses the CLIP embedding of the straightened object crop as its query. Four eight-head attention blocks, an MLP and attention pooling produce a 512-dimensional vector. We select the candidate with the highest dot product against that vector. For a single detection we use the full image.

**Associative retrieval.** The supplied memory contains 13 median context vectors, each with 512 entries. We use the selected context embedding and the CLIP class-name embedding to query the same memory. Each query produces a softmax-weighted average of the stored vectors with beta = 8. A cosine similarity at or below 0.90 flags the detection. This threshold comes from the original demo script.

## 4. Results and Observations

We ran both settings on the same 20 validation images on CPU, with YOLO confidence set to 0.25. We matched predictions to labels by class and rotated polygon IoU of at least 0.50. Each ground-truth box could be matched once. We included size 256 because it is close to the original training setting and divisible by 32. The submission script defaults to 256 based on these results.

| YOLO size | Detections | Matched | Precision | Recall | Flagged |
|---|---|---|---|---|---|
| 896 | 39 | 3 | 7.7% | 13.6% | 19 |
| 256 | 22 | 16 | 72.7% | 72.7% | 7 |

Precision and recall above refer to object detection at the stated confidence and IoU settings.

**Input 896.** Context similarity ranged from 0.7429 to 0.9999, with a median of 0.9418. Selected views by count: column 4, row 3, box2x 2, full 30.

**Input 256.** Context similarity ranged from 0.7031 to 0.9998, with a median of 0.9675. Selected views by count: column 1, row 0, box2x 0, full 21.

The validation set was also available to the detector training run for checkpoint selection. These measurements therefore describe the supplied validation split. Any object missed by YOLO is also missed by the context stage.

## 5. Example Output




Figure 1. Archived overlay image at YOLO size 256. Green boxes pass the context cutoff. Red boxes are flagged.

| Predicted class | Selected view | YOLO confidence | Similarity |
|---|---|---|---|
| knife | box2x | 0.447 | 0.8651 |
| pen | box2x | 0.408 | 0.8457 |

## 6. Limitations and Conclusion

The pipeline runs from a single Python file and produces box overlays and a JSON file with a score for each detection. Our validation run shows that the detector misses objects and makes incorrect proposals, both of which affect the context stage. The 0.90 cutoff is uncalibrated. Measuring anomaly precision and recall would require a separate set of scenes with normal and anomalous objects labelled.

## 7. Base Paper and References

**Base paper.** H. Ramsauer et al., "Hopfield Networks is All You Need," ICLR, 2021. We used its modern Hopfield update for memory retrieval. arXiv:2008.02217.

A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML, 2021. PMLR 139. Ultralytics, Oriented Bounding Box Detection.